

Normal Blood Pressure Control and the Evaluation of Hypertension

William J. Elliott

NORMAL BLOOD PRESSURE CONTROL

Systemic arterial blood pressure (BP) is the pressure of the blood within the arteries exerted against the arterial wall and is produced by the contraction of the left ventricle (producing blood flow) and the resistance of the arteries and arterioles. *Systolic* blood pressure (SBP), or *maximum* BP, occurs during left ventricular systole. *Diastolic* blood pressure (DBP), or *minimum* BP, occurs during ventricular diastole. The difference between SBP and DBP is the pulse pressure.¹ The mean arterial pressure (MAP) is calculated as the DBP plus one-third of the pulse pressure.

Blood flow (Q), as defined by the hydraulic analogy of Ohm's law, varies directly with the change in pressure (P), across a blood vessel, and varies inversely with the resistance (R), defined as $Q = P/R$. Rearrangement shows that pressure varies directly with blood flow and resistance: $P = QR$. Ohm's law suffices for an overall view of the circulation. However, the flow within a vessel is governed by the Hagen-Poiseuille equation:

$$Q = \Delta P \times \left(\frac{\pi r^4}{8L} \right) \times (1/\eta)$$

where r is the radius of the pipe, L is its length, and η is the coefficient of viscosity. Thus, as the lumen of a vessel decreases, the pressure increases by the fourth power of the radius to maintain the same blood flow.

Normal BP is controlled by cardiac output and the total peripheral resistance and is dependent on the heart, the blood vessels, the extracellular volume, the kidneys, the nervous system, humoral factors, and cellular events at the membrane and within the cell (Fig. 34.1). Cardiac output is determined by the stroke volume in liters per minute (L/min) and the heart rate. In turn, stroke volume is dependent on intravascular volume (regulated by the kidneys) and myocardial contractility. Myocardial contractility involves sympathetic and parasympathetic control of heart rate, intrinsic activity of the cardiac conduction system, complex membrane transport and cellular events requiring influx of calcium that lead to myocardial fiber shortening and relaxation, and effects of humoral substances (e.g., catecholamines) on increasing heart rate and myocardial fiber tension.

Total peripheral resistance is regulated by baroreflexes and sympathetic nervous system (SNS) activity, response to neurohumoral substances and endothelial factors, myogenic responses, and intercellular events mediated by receptors and mechanisms for signal transduction.² Baroreflexes are derived from (1) high-pressure baroreceptors

in the aortic arch and carotid sinus and (2) low-pressure cardiopulmonary baroreceptors in ventricles and atria. Aortic baroreceptor nerve fibers travel via the vagus nerve (cranial nerve X); carotid sinus fibers travel via the glossopharyngeal nerve (cranial nerve IX). These receptors respond to stretch (high pressure) or filling pressures (low pressure) and send tonic inhibitory signals to the brainstem. If BP and tonic inhibition increase, inhibition of sympathetic efferent outflow occurs, decreasing vascular resistance and heart rate. However, if BP decreases, less tonic inhibition occurs, and heart rate and peripheral vascular resistance (PVR) increase, thereby increasing BP.

The brainstem cardiovascular (CV) centers are localized in the dorsomedial medulla. Neural afferents from cranial nerves IX and X are integrated in the nucleus tractus solitarius (NTS). From here, vasoconstriction and increased heart rate are mediated through the caudal and rostral ventrolateral medulla by the SNS. Efferents from the NTS communicate with the nucleus ambiguus (vagal nucleus) to decrease heart rate via the vagus nerve. Also, the central neural control of kidney function modulates renal blood flow, glomerular filtration rate (GFR), excretion of sodium and water, and renin release. These factors in turn regulate intravascular volume, vascular resistance, and BP.³ This complex physiology has taken on greater clinical relevance recently, given the mixed results of pivotal trials of percutaneous, catheter-based renal denervation.^{4,5}

Inhibitory reflexes also originate in the kidney. Increases in urine flow rate increase renal pelvic pressure, which stretches the renal pelvic wall and leads to activation of mechanosensory nerves in the pelvic wall. Activation of these sensory nerves decreases renal sympathetic nerve activity and induces diuresis and natriuresis, an inhibitory renorenal reflex response. The responsiveness of the renal sensory nerves is modulated by dietary sodium. A high sodium intake enhances the responsiveness of the afferent renal mechanosensory nerves; conversely, renal denervation may increase urinary flux and fractional excretion of sodium,⁶ although not all recent studies agree.

Numerous vasoactive substances have effects on blood vessels, the heart, the kidneys, and the central nervous system (CNS) to regulate BP (Table 34.1). The renin-angiotensin-aldosterone system (RAAS) regulates volume and PVR (Fig. 34.2), particularly in the long term (hours to weeks; Fig. 34.3). Angiotensin II (Ang II) constricts vascular smooth muscle; stimulates aldosterone secretion; potentiates SNS activity; stimulates salt and water reabsorption in the proximal tubule; stimulates prostaglandin, nitric oxide (NO), and endothelin release; increases thirst; and stimulates vascular remodeling and inflammation. Aldosterone stimulates sodium channels in distal renal tubular

Some Factors Involved in the Regulation of Blood Pressure

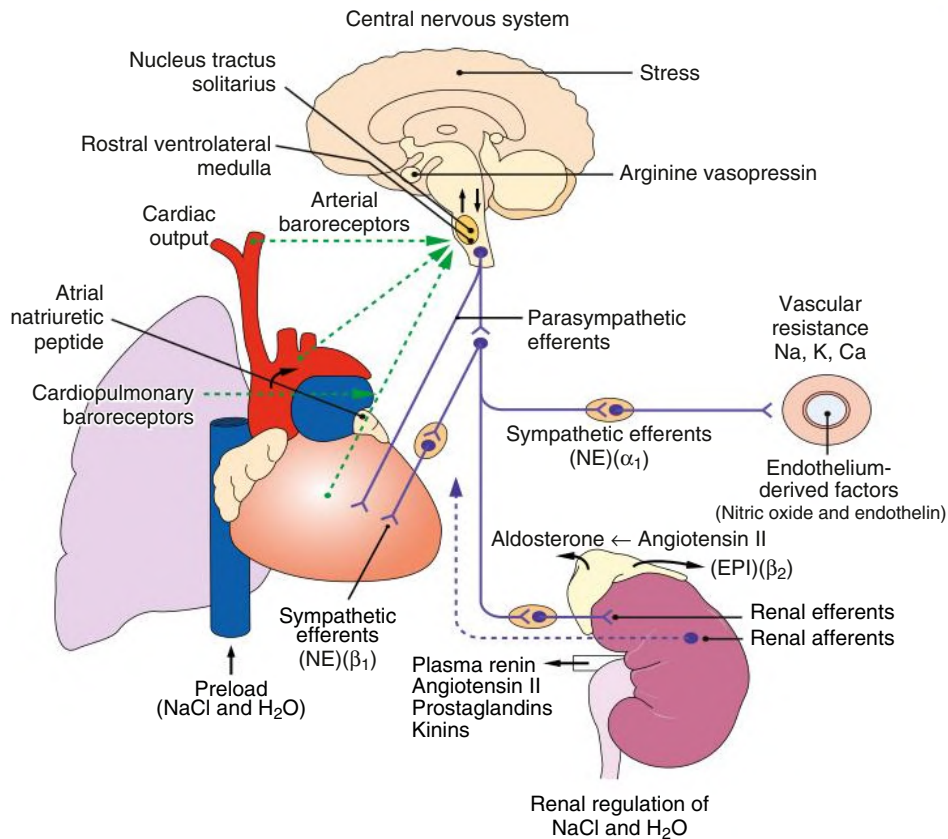


Fig. 34.1 Some Factors Involved in Regulation of Blood Pressure. Dashed lines refer to feedback to the central nervous system from cardiovascular (green dash) or renal (purple dash) sites. EPI, Epinephrine; NE, norepinephrine.

epithelium, leading to sodium retention and potassium excretion. Novel aldosterone antagonists have antiinflammatory and other beneficial effects in vascular cells, heart, and kidneys.⁷ Plasma concentrations of renin and aldosterone are both inversely related to salt intake and are influenced by many antihypertensive medications.

The second major effector system, working primarily over seconds to minutes (see Fig. 34.3), is the SNS. Sympathetic nerve endings release the vasoconstrictor (norepinephrine) that binds the α -adrenergic receptor (adrenoceptor) on vascular cells, renal cells, and other cells (e.g., adipocytes). Epinephrine increases heart rate, stroke volume, and SBP through α - and β -adrenoceptors. The hormone is released from the adrenal medulla. Increased sympathetic tone has long-term influences on CV regulation and may cause hypertension and contribute to chronic kidney disease (CKD).⁸ In the kidneys, sympathetic nerves mediate renin release. Furthermore, innervation of each individual nephron affects sodium reabsorption. In doing so, the SNS regulates both effective circulating fluid volume and PVR.

The kallikrein-kinin system counters the RAAS and produces vasodilator kinins, which stimulate prostaglandin and NO production (see Fig. 34.2). Prostaglandin E and prostacyclin block the vasoconstriction by Ang II and norepinephrine. Two endothelium-derived factors have opposite effects on the blood vessels: NO is a vasodilator, whereas the endothelins are vasoconstrictors. Natriuretic peptides, of the atrial, brain, or C-types, induce vasodilation and natriuresis and inhibit other vasoconstrictors (RAAS, SNS, endothelin). Chapter 37 discusses all the

drugs that affect the previously discussed physiologic systems that regulate BP.

Other physiologic systems that control BP can be used as investigative tools but not yet therapeutically in humans. Renalase is a flavin adenine dinucleotide-dependent amine oxidase that is secreted by the kidney, circulates in the blood, and modulates cardiac function and systemic BP by metabolizing catecholamines; there is debate about whether mutations in the gene coding for renalase are associated with hypertension.⁹ Heme oxygenase-1 is an intrarenal and systemic BP modulator that inhibits oxidants, resulting in decreased BP; in one study, individuals with an unfavorable gene promoter for this enzyme had a higher prevalence of hypertension and increased mortality.¹⁰ Endogenous digitalis-like factors, which inhibit cell surface Na⁺,K⁺-ATPase and include an ouabain-like factor and marinobufagenin, appear to regulate BP, CV, and renal function.¹¹ Urotensin II is a locally expressed vasoconstrictive cyclic vasoactive peptide that stimulates proliferation of vascular smooth muscle cells and fibroblasts, inhibits insulin release, and modulates GFR. Nonetheless, high plasma urotensin II levels were associated with reduced CV complications and death in patients with CKD, stages 2 to 5.¹² In obese persons, leptin may increase BP by activating the CNS through a melanocortin pathway.¹³ Although its clinical importance is still debated, higher serum uric acid levels have been associated with BP and CV outcomes in many (but not all) cohort studies, a mendelian randomization analysis, and meta-analysis of 15 clinical trials¹⁴; at least two hypouricemic drugs

TABLE 34.1 Some Vasoactive Substances That Modulate Blood Pressure

Group	Compound	Cellular Effects
Catecholamines	Norepinephrine, epinephrine, dopamine	Adrenergic receptors (α_1 , α_2 , β_1 , β_2) causing protein phosphorylation and increased intracellular calcium through G proteins linked to ion channels or second messengers (cyclic nucleotides, phosphoinositide hydrolysis)
RAS	Angiotensin II	Angiotensin receptors (AT_1 , AT_2 , AT_4) cause increased intracellular calcium and protein phosphorylation through second messenger, phosphoinositide hydrolysis, and activated protein kinases Angiotensin II stimulates aldosterone
Mineralocorticoids	Aldosterone	<i>Genomic</i> : binds to cytoplasmic mineralocorticoid receptor, translocates to nucleus, modulates gene expression, and signal transduction and effectors ($S_{\beta}K$, CHIF, K_r -Ras), which increases transport proteins (increasing ENaC number and open probability) <i>Nongenomic</i> : effects through separate membrane or cytosolic proteins
Kallikrein-kinin system	Bradykinin	Bradykinin receptors (B_1 , B_2), B_2 -G protein coupling causes activation of phospholipase C, increased inositol phosphates, and intracellular calcium
Arachidonic acid oxidation products	Prostaglandins: PGE, prostacyclin, thromboxanes Lipoxygenase enzyme products: leukotrienes, hydroxyeicosatetraenoates	Nine prostaglandin receptors coupled to G proteins (e.g., PGI_2 [receptor IP], PGE_2 [receptors EP ₁ , EP ₂]); $PGF_{2\alpha}$ (receptor FP)
Endothelium-derived factors	Endothelium-derived relaxing factor (nitric oxide) Endothelins (ET-1, ET-2, ET-3)	Increased levels of cyclic guanosine monophosphate cause activation of protein kinases G proteins activate phospholipase C and L-type calcium channels. Class 2 G protein-coupled receptor
Natriuretic peptides	Atrial, brain, and C-type	Activation of three receptor types; further effects mediated by cGMP
Posterior pituitary hormones	Arginine vasopressin	Vasopressin receptors (AVPR 1A; AVPR 1B) mediated by second messenger system, phosphatidylinositol/calcium; AVPR2 effects via adenylate cyclase (cAMP)
Cyclic vasoactive peptides	Urotensin II	Binds to G protein receptor GPR 14
Other substances	Acetylcholine, adenosine, insulin, neuropeptide Y, serotonin, sex hormones (estrogens, progesterone, androgens), glucocorticoids, other mineralocorticoids, substance P, vasopressin, reninase, heme oxygenase 1, uric acid	

Ang II, Angiotensin II; *AVPR*, arginine vasopressor receptor; *cAMP*, cyclic adenosine monophosphate; *cGMP*, cyclic guanosine monophosphate; *ENaC*, amiloride-sensitive epithelial sodium channel; *PGE*, prostaglandin E; *RAS*, renin-angiotensin system; *UT II*.

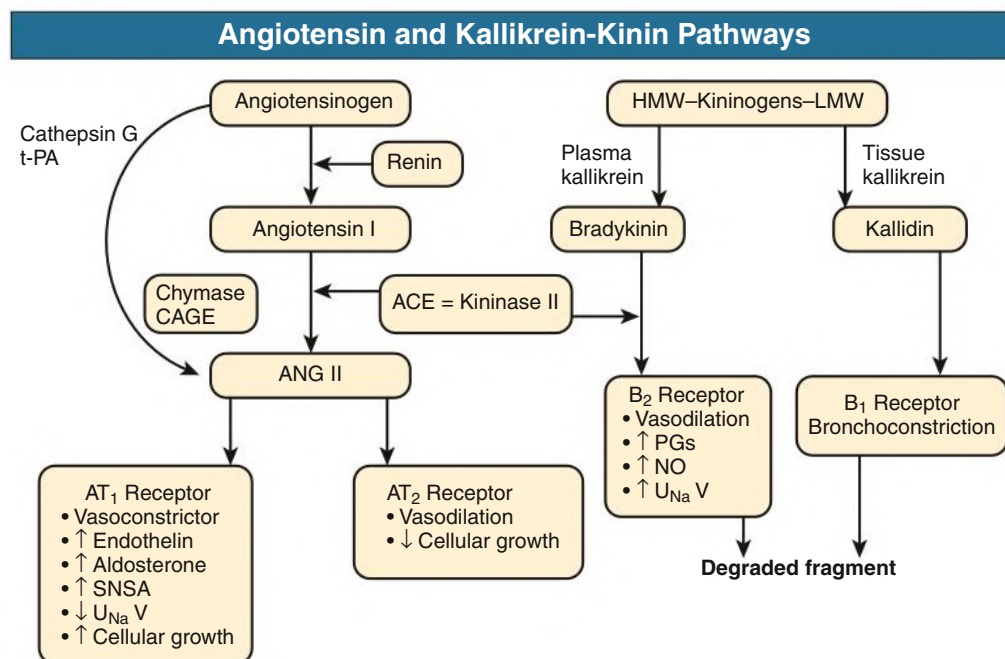


Fig. 34.2 Interactions and functions of renin-angiotensin and kallikrein-kinin systems. *ACE*, Angiotensin-converting enzyme; *ANG II*, angiotensin II; *AT₁*, *AT₂*, angiotensin receptors; *B₁*, *B₂*, bradykinin receptors; *CAGE*, chymostatin-sensitive angiotensin II-generating enzyme; *HMW*, high molecular weight; *LMW*, low molecular weight; *NO*, nitric oxide; *PGs*, prostaglandins; *SNSA*, sympathetic nervous system activity; *t-PA*, tissue plasminogen activator; *UNaV*, urinary sodium excretion.

Temporal Sequence for Adjustment of Blood Pressure Control

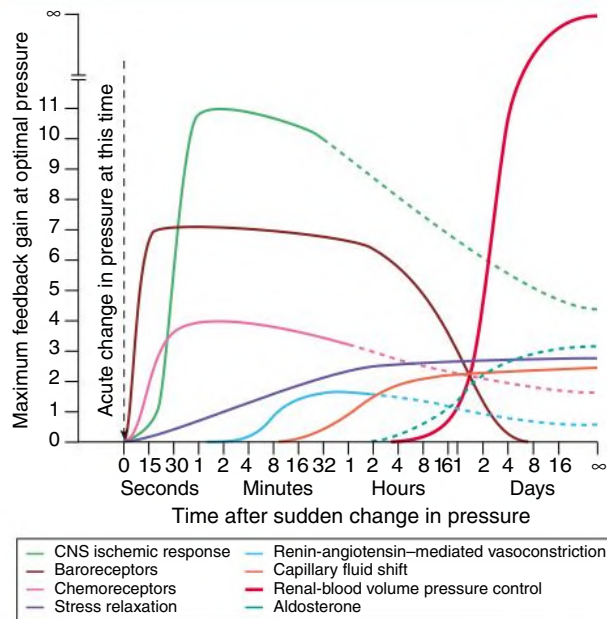


Fig. 34.3 Temporal Sequence for Adjustment of Blood Pressure Control. Degree of activity, expressed as feedback gain, of several arterial blood pressure (BP) control systems at various times after a sudden change in arterial BP. Note the infinite gain of the kidney volume mechanism for BP control. CNS, Central nervous system. (From Hall JE. Role of the kidneys in long-term control of arterial pressure and in hypertension: the integrated system for arterial pressure regulation. In: Hall J, Hall M. *Guyton and Hall's Textbook of Medical Physiology*. 14th ed. Philadelphia: Elsevier; 2020:229–244.)

have lowered BP in pilot studies. Intracellular urate also may activate the renin-angiotensin system (RAS), induce intrarenal oxidative stress, block endothelial NO, and directly affect the vasculature.¹⁵ T cells also may be involved in primary hypertension, possibly via heat shock protein 70 and/or interleukin-17, and are present in kidneys and blood vessels where they release Ang II and oxidants.¹⁶ Chemoreceptors in the brain medulla and the carotid and aortic bodies respond to changes in carbon dioxide and oxygen tension, resulting in renal vasoconstriction and dilation of the CNS and coronary vasculature; these phenomena may be reversible after device-based therapy.¹⁷ The small guanosine triphosphatase, Rho, and its kinase stimulate vasoconstriction and may have a role in cerebral artery spasm, hypertension, heart failure, and other CV conditions; an inhibitor Rho kinase, netarsudil, was recently FDA-approved for ocular hypertension.¹⁸

The kidney is recognized as a source of hypertension, based on renal cross-transplantation experiments¹⁹ and the hypotensive effects of bilateral nephrectomy in patients with end-stage kidney disease (ESKD) and severe hypertension. Activation of CNS and kidney SNS pathways are important in long-term BP control, in part by delayed effects on the kidneys.²⁰ Renal sympathetic denervation by radiofrequency ablation through the renal arteries has led to substantial reduction in BP in patients with resistant hypertension, but a large trial did not achieve significantly better BP control at 6 months, compared with sham-operated controls.^{21,22} This negative result more likely stemmed from technical issues in the conduct of the trial rather than indicating that the SNS does not contribute to the pathophysiology of hypertension.

DEFINITION AND CLASSIFICATION OF HYPERTENSION

In the general population, casual or in-office BP follows an approximately normal distribution, with a shift to the left in the young and the opposite in older people or those with CKD. Thus, any definition of hypertension is arbitrary. Hypertension is often asymptomatic, with symptoms more commonly attributed to sequelae of hypertension or its treatment. Hypertension may be classified by its associated morbidity and mortality, as increases over arbitrary cut points, or by thresholds defining therapeutic benefit.

Blood Pressure Classification by Link to Future Morbidity and Mortality

The first approach defines hypertension by relating BP levels to the risk for morbidity and mortality. The association of SBP and DBP with CV and kidney complications is continuous over the entire BP range.²³ Death from both heart disease and stroke increases progressively and linearly from BP as low as 115/75 mm Hg upward in all age groups from 40 to 89 years (Figs. 34.4 and 34.5). In observational studies an increase in SBP of 20 mm Hg or DBP of 10 mm Hg was associated with a doubling of mortality from heart disease or stroke. Several interventional trials have observed a J-shaped curve (suggesting that mortality increases below a threshold SBP), yet this phenomenon is also seen in the bottom decile in both individual and pooled observational studies of untreated subjects (see Figs. 34.4 and 34.5), indicating that those in the lowest decile have a higher risk, irrespective of treatment. Elevated SBP has been consistently identified by the Global Burden of Disease study to be the leading contributor to disability-adjusted life years worldwide. The investigators characterized an SBP between 110 and 115 mm Hg as the “theoretical minimum risk level” for BP. Although all national and international hypertension guidelines agree about the continuous relationship between risk and BP, as well as a recommendation for lifestyle modifications in people with BP 120/80 mm Hg or greater, the thresholds and categories have continued to evolve. The Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC 7) introduced the term *prehypertension* for individuals with SBPs from 120 to 139 mm Hg or DBPs between 80 and 89 mm Hg (Table 34.2),²⁴ which was retained by the American Society of Hypertension/International Society of Hypertension 2013 guidelines and by the Expert Panel appointed to JNC 8.²⁵ Somewhat different terms were used by the 2018 European Society of Hypertension/European Society of Cardiology guidelines (Table 34.3)²⁶ and the 2020 International Society of Hypertension guidelines (Table 34.4).²⁷ A completely different approach (independent of specific BP thresholds) was taken by the Writing Group of the American Society of Hypertension (WG-ASH), which proposed that hypertension is a complex CV disorder that included target organ damage, early disease biomarkers (including BP), and CV risk factors²⁸ (Table 34.5).

Blood Pressure Classification by Population-Based Percentile

A second approach defined hypertension by the frequency distribution within a population. Until 2017, hypertension in American children was determined on a purely statistical basis: SBP or DBP at or above the 95th percentile for age, sex, and height, measured on at least three occasions.²⁹ In this population, age, sex, body size, and race/ethnicity are all significant and strong predictors of BP. Using a frequency distribution method is less helpful for identifying a threshold for initiation of antihypertensive treatment but is common in epidemiologic studies. The crude prevalence of hypertension in adults (older than 20 years)

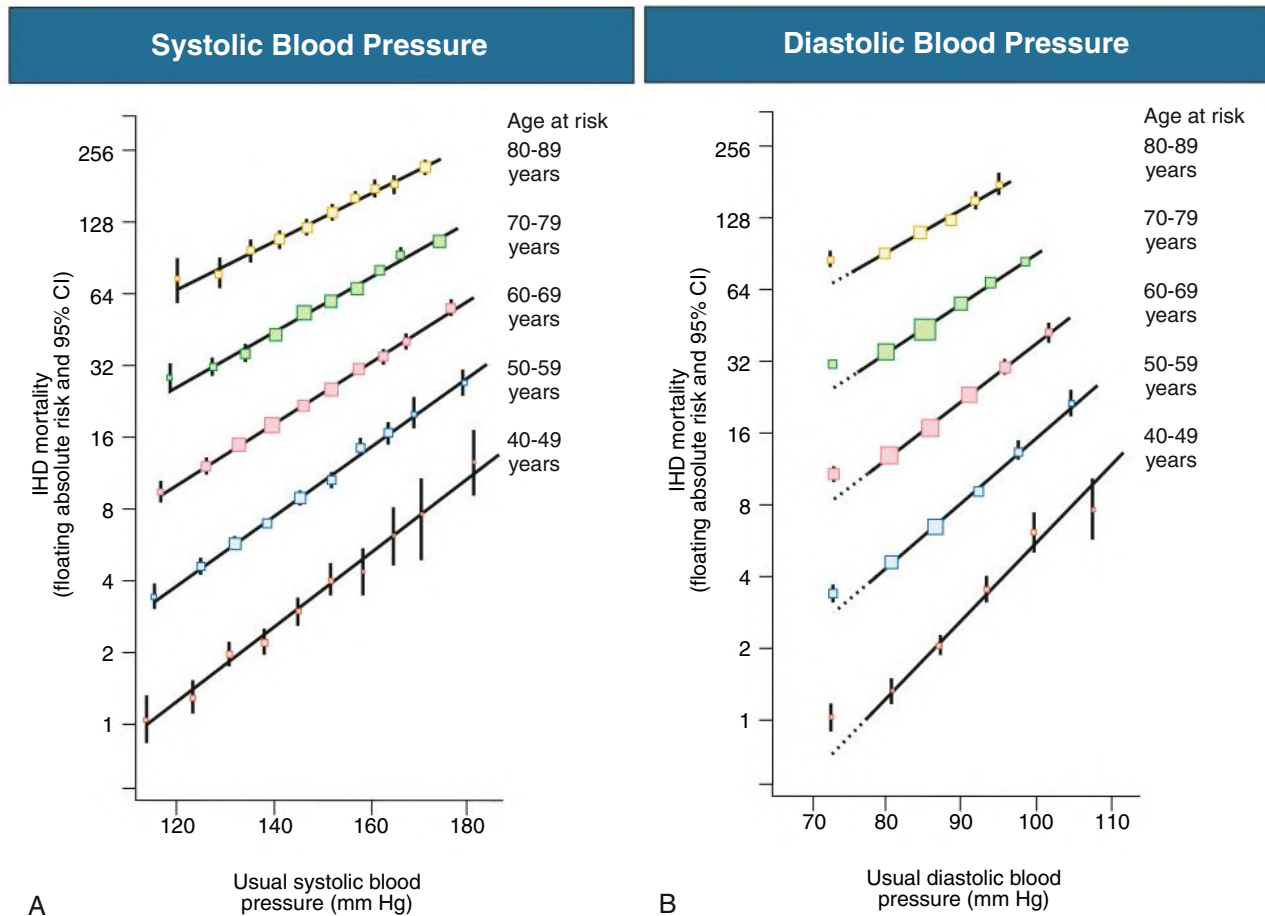


Fig. 34.4 Ischemic Heart Disease Mortality Rate in Each Decade of Age Versus Usual Blood Pressure at Start of That Decade. (A) Systolic blood pressure. (B) Diastolic blood pressure. Floating absolute risk is a relative risk score that adjusts for the absolute death rate within a particular age group. The size of the squares correlates inversely with the variance of the data collected for that data point. *CI*, Confidence interval; *IHD*, ischemic heart disease. (From Lewington S, Clarke R, Qizilbash N, et al. Prospective studies collaboration: age-specific relevance of usual blood pressure to vascular mortality—a meta-analysis of individual data for one million adults in 61 prospective studies. *Lancet*. 2002;360:1903–1913.)

in the United States, traditionally defined as BP of 140/90 mm Hg or greater or taking antihypertensive medication, has increased progressively from 11% in 1939 to 33.5% in 2013 to 2014, but has been relatively stable since the mid-1990s ($31\% \pm 2\%$, after age adjustment).³⁰

Blood Pressure Classification by Threshold of Therapeutic Benefit

The third concept for defining hypertension is derived from randomized controlled trials (RCTs) that compared two or more BP targets and showed significant reductions in morbidity and mortality. The Hypertension Optimal Treatment (HOT) study showed no significant outcome differences among 18,790 hypertensive subjects randomized to DBP targets of 80, 85, or 90 mm Hg or less, although a significant benefit was seen in diabetic patients with a diastolic target of less than 80 mm Hg.³¹ A post hoc analysis of the in-trial BPs determined that maximal prevention of major CV events was seen at a BP of 138.5/82.6 mm Hg. Of the relatively few outcome-based RCTs comparing traditional and lower BP targets, only the Systolic Blood Pressure Intervention Trial (SPRINT) showed significant reductions in both CV events and mortality when treating to a SBP target of less than 120 compared with the traditional less than 140 mm Hg.³² Some have argued that this benefit came at a huge (and largely unspecified) cost of more medications, more office visits, more adverse effects

(including emergency department visits), and greater than 30% worsening of renal function in those without CKD at baseline. Others have cast doubt about SPRINT's generalizability, as less than 17% of adult Americans with hypertension would have been eligible to participate because of the large number of exclusion criteria (which included diabetes mellitus, prior stroke, polycystic kidney disease, or estimated GFR <20 mL/min/1.73 m²). A network meta-analysis then combined the SPRINT results with those of 41 other randomized clinical trials, using *achieved* in-trial SBPs in 144,220 subjects as the independent variable. For both cardiovascular events and all-cause mortality, the lowest risk was seen in subjects with SBPs between 120 and 124 mm Hg, significantly lower than those with SBPs between 130 and 134 mm Hg.³³ Three other independent meta-analyses also concluded that major adverse cardiovascular events were significantly reduced in subjects treated to “lower SBPs” (typically <130 mm Hg) compared with “higher SBPs” (typically ≥ 140 mm Hg).

These data resulted in the 2017 ACC/AHA United States hypertension guideline.³⁴ This guideline “redefined” hypertension as a usual office BP at or below 130/80 mm Hg, added the new diagnostic category of “elevated blood pressure,” reemphasized the value of out-of-office BP measurements, reinforced lifestyle modifications as appropriate for most people with higher than normal BPs, and revised some treatment targets for specific patient groups (see Table 34.6 for

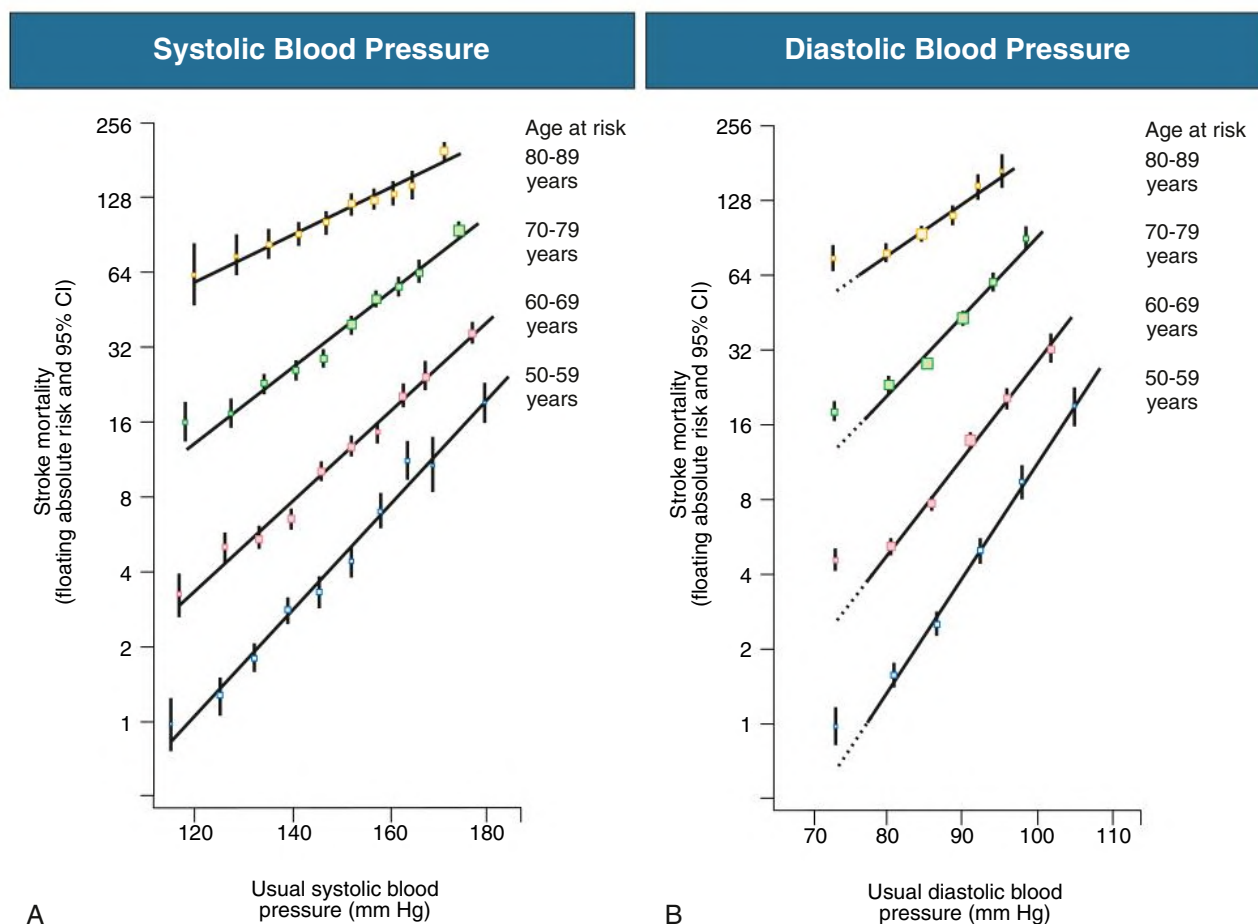


Fig. 34.5 Stroke Mortality Rate in Each Decade of Age Versus Usual Blood Pressure (BP) at Start of That Decade. (A) Systolic BP. (B) Diastolic BP. Floating absolute risk is a relative risk score that adjusts for the absolute death rate within a particular age group. The size of the squares correlates inversely with the variance of the data collected for that data point. *CI*, Confidence interval. (From Lewington S, Clarke R, Qizilbash N, et al. Prospective studies collaboration: age-specific relevance of usual blood pressure to vascular mortality—a meta-analysis of individual data for one million adults in 61 prospective studies. *Lancet*. 2002;360:1903–1913.)

TABLE 34.2 JNC 7 Classification of Blood Pressure for Adults (2003)

Classification	SBP (mm Hg)		DBP (mm Hg)
Normal	<120	and	<80
Prehypertension	120–139	or	80–89
Stage 1 hypertension	140–159	or	90–99
Stage 2 hypertension	≥160	or	≥100

DBP, Diastolic blood pressure; JNC, Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure; SBP, systolic blood pressure.

more details). The 2017 US hypertension guideline of the American College of Cardiology/American Heart Association (ACC/AHA) is controversial; some societies have declined to endorse or adopt it. About 3 months earlier, the American Academy of Pediatrics released its Clinical Practice Guideline for Screening and Management of High Blood Pressure in Children and Adolescents²⁹; the authors of that pediatric guideline were privy to an advance copy of the 2017 adult

guideline and arranged their recommendations to be consistent with it. These new definitions have increased the overall prevalence of hypertension in US adults to 46% but decreased the prevalence in children and adolescents to 3.5%. The ACC/AHA 2017 US hypertension guideline also recommended an office BP target below 130/80 mm Hg for individuals with a 10-year cardiovascular risk greater than 10%, but retained the traditional less than 140/90 mm Hg for most lower-risk individuals.

The most recent set of hypertension guidelines from the Kidney Disease: Improving Global Outcomes (KDIGO) did not formally redefine hypertension but recommended that only “standardized office BPs,” supplemented by out-of-office BP measurements, be used in clinical decision-making, and (based on “weak” evidence) that all patients with chronic kidney disease be treated to a systolic BP less than 120 mm Hg.³⁵ The authors concluded that this target may be harmful if “standardized” BPs are not measured.

The concept of evaluating a person’s “total CV risk” had been prominent in several prior US guidelines on BP, dyslipidemia, and acute coronary syndrome. It was largely abandoned by JNC 7 but still plays a major role in other countries^{26,27,36} and was incorporated in the 2017 ACC/AHA US hypertension guideline.³⁴ Age, sex, and ethnicity

TABLE 34.3 European Society of Cardiology and European Society of Hypertension Classification Scheme and Diagnostic Thresholds for Hypertension (2018)

CLASSIFICATION OF OFFICE BP				THRESHOLD BPS FOR DIAGNOSIS OF HYPERTENSION			
Category	Systolic (mm Hg)		Diastolic (mm Hg)	Category	Systolic (mm Hg)		Diastolic (mm Hg)
Optimal	<120	and	<80	Office BP	≥140	and/or	≥90
Normal	120–129	and/or	80–84	Ambulatory BP			
High normal	130–139	and/or	85–89	Daytime (or awake)	≥135	and/or	≥85
Grade 1 hypertension	140–159	and/or	90–99	Nighttime (or asleep)	≥120	and/or	≥70
Grade 2 hypertension	160–179	and/or	100–109	24-hour	≥130	and/or	≥80
Grade 3 hypertension	≥180	and/or	≥110	Home BP	≥135	and/or	≥85
Isolated systolic hypertension	≥140	and/or	<90				

Modified from Williams B, Mancia G, Spierling W, et al. for the Task Force for the Management of Arterial Hypertension of the European Society of Cardiology and the European Society of Hypertension. 2018 ESC/ESH guidelines for the management of arterial hypertension. *J Hypertens*. 2018;36:2284–2309.

TABLE 34.4 International Society of Hypertension Classification Scheme and Diagnostic Thresholds for Hypertension (2020)

CLASSIFICATION OF OFFICE BP				THRESHOLD BPS FOR DIAGNOSIS OF HYPERTENSION			
Category	Systolic (mm Hg)		Diastolic (mm Hg)	Category	Systolic (mm Hg)		Diastolic (mm Hg)
Normal BP	<130	and	<85	Office BP	≥140	and/or	≥90
High-normal BP	130–139	and/or	85–89	Office BP	≥140	and/or	≥90
Grade 1 hypertension	140–159	and/or	90–99				
Ambulatory BP							
Grade 2 hypertension	≥160	and/or	≥100	24-hour average	≥130	and/or	≥80
				Daytime (or awake)	≥135	and/or	≥85
				Nighttime (or asleep)	≥120	and/or	≥70
				Home BP	≥135	and/or	≥85

BP, Blood pressure.

From Unger T, Borghi C, Charchar F, et al. 2020 International Society of Hypertension global hypertension practice guidelines. *Hypertension*. 2020;75:1334–1357.

TABLE 34.5 Working Group: American Society of Hypertension Definition and Classification of Hypertension

Class	BP Elevation		CV Disease ^a	CV Risk Factors	Early Disease Markers	Target Organ Disease
Normal	Normal or rare	or	None	None or few	None	None
Hypertension						
Stage 1	Occasional intermittent	or	Early	Several	Usually present	None
Stage 2	Sustained	or	Progressive	Many	Overtly present	Early signs present
Stage 3	Marked and sustained	or	Advanced	Many	Overtly present with progression	Overtly present with or without CV disease events

^aCardiovascular (CV) disease determined by constellation of risk factors, early disease markers, and target organ disease.

BP, Blood pressure.

Modified from Giles TD, Materson BJ, Cohn JN, Kostis JB. Definition and classification of hypertension: an update. *J Clin Hypertens (Greenwich)*. 2009;11:611–614. Correction *J Clin Hypertens (Greenwich)*. 2010;12:13.

are important nonmodifiable risk factors, whereas low-density lipoprotein (LDL) cholesterol, smoking, control of diabetes, obesity, and left ventricular hypertrophy are potentially modifiable³⁶ (Fig. 34.6). The cluster of risk factors that increase CV risk and are often associated with hypertension is termed the *metabolic syndrome*³⁷ (Table 34.7).

CKD, as defined by decreased GFR or by increased urinary albumin excretion, is also recognized as an independent risk factor for ESKD, CV events, and death.³⁵ JNC 7 included recommendations for serial follow-up evaluation of BP and CV risk factors based on initial BP measurement (Table 34.8).

TABLE 34.6 ACC/AHA 2017 US Hypertension Guideline: Categories of Blood Pressure in Adults (mm Hg)

Category	Systolic		Diastolic
Normal	<120 mm Hg	and	<80 mm Hg
Elevated	120–129 mm Hg	and	<80 mm Hg
Hypertension			
Stage 1	130–139 mm Hg	or	80–89 mm Hg
Stage 2	≥140 mm Hg	or	≥90 mm Hg

Individuals with systolic and diastolic blood pressures in two different categories should be assigned into the higher category. Blood pressures are based on an average of ≥ 2 careful readings obtained on ≥ 2 occasions, usually separated by a week or more.

ACC/AHA, American College of Cardiology/American Heart Association. From Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: a report of the American College of Cardiology/American Heart Association Task Force on clinical practice guidelines. *Hypertension*. 2018; 71:e13–e115.

Special Definitions

Prehypertension. Although no longer a feature of the ACC/AHA 2017 US hypertension guideline,³⁴ prehypertension was defined in JNC 7 as SBP of 120 to 139 and DBP of 80 to 89 mm Hg.²⁴ Prehypertension was found in 28% of American adults in the US National Health and Nutrition Examination Survey (NHANES) 2005 to 2006.³⁸ Prehypertension is associated with age, obesity, dyslipidemia, impaired fasting glucose levels, a higher risk for both CV and renal events, and progression to hypertension (which was significantly delayed by drug therapy in two RCTs). Lifestyle modifications (see [Chapter 36](#)) are recommended by essentially all guidelines for affected people until the BP meets or exceeds levels corresponding to hypertension.

“White coat” hypertension. “White coat” hypertension, defined as normal BP during usual daily activities yet elevated only in a clinical setting, has a prevalence of 20% to 25% in hypertensive persons and is most accurately diagnosed by ambulatory BP monitoring (with home BP measurements a distant second choice). Unfortunately, ambulatory BP monitoring is not widely available for routine clinical use in many countries (including the United States) despite having become the gold standard for diagnosis in research settings. The white coat (or “alerting”) phenomenon is less common when a nurse or technician measures the BP (compared with when taken by a physician), and is

Absolute Risk of Cardiovascular Disease by Systolic Blood Pressure

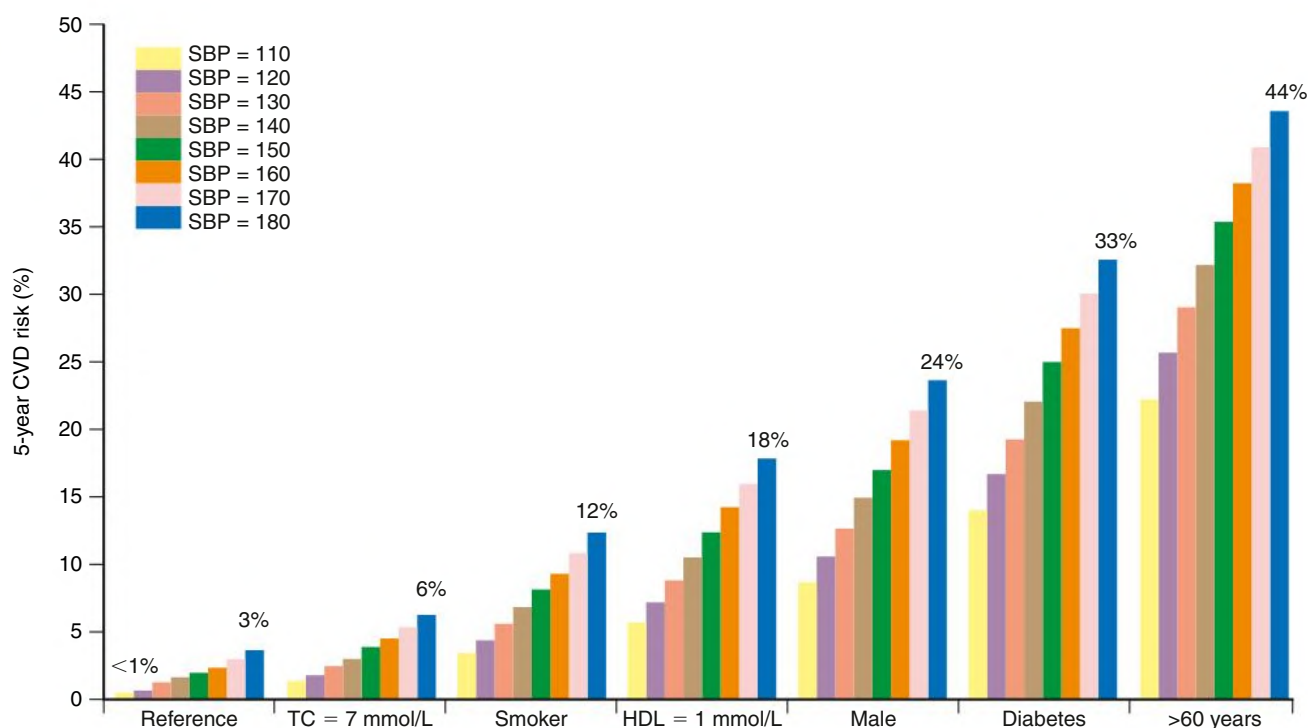


Fig. 34.6 Absolute Risk for Cardiovascular Disease During 5 Years in Patients by Systolic Blood Pressure at Specified Levels of Other Risk Factors. Reference category is a nondiabetic, nonsmoking woman age 50 years with total cholesterol (TC) level of 4 mmol/L (155 mg/dL) and high-density lipoprotein (HDL) level of 1.6 mmol/L (62 mg/dL). Risks are given for systolic blood pressure (SBP) levels of 110, 120, 130, 140, 150, 160, 170, and 180 mm Hg. In the other categories, additional risk factors are added consecutively; for example, the diabetes category is a diabetic 50-year-old male cigarette smoker with TC level of 7 mmol/L (270 mg/dL) and HDL level of 1 mmol/L (39 mg/dL). CVD, Cardiovascular disease. (Modified from Jackson R, Lawes CM, Bennett A, et al. Treatment with drugs to lower blood pressure and blood cholesterol based on an individual's absolute cardiovascular risk. *Lancet*. 2005;365:434–441.)

TABLE 34.7 Common Definitions for Metabolic Syndrome

Criterion	NCEP ATP III, 2005 Update (≥3 Criteria)	International Diabetes Federation, 2005 (Obesity + 2 Other Criteria)
Abdominal obesity	Waist circumference	Required for diagnosis
Men	>40 inches (>102 cm)	>94 cm
Women	>35 inches (>88 cm)	>80 cm
Hypertriglyceridemia	>150 mg/dL (>1.7 mmol/L)	>150 mg/dL (>1.7 mmol/L) or treatment
Low HDL		
Men	<40 mg/dL (<1.03 mmol/L)	<40 mg/dL (<1.03 mmol/L) or treatment
Women	<50 mg/dL (<1.30 mmol/L)	<50 mg/dL (<1.30 mmol/L) or treatment
Hypertension	≥130/85 mm Hg or taking antihypertensive medication	≥130/85 mm Hg or taking antihypertensive medication
Impaired fasting glucose or diabetes	Glucose ≥100 mg/dL (≥5.6 mmol/L) or taking insulin or hypoglycemic medication	Glucose ≥100 mg/dL (≥5.6 mmol/L) or taking insulin or hypoglycemic medication

HDL, High-density lipoprotein; NCEP ATP III, National Cholesterol Education Program—Adult Treatment Panel III.

Modified from Jackson R, Lawes CM, Bennett A, et al. Treatment with drugs to lower blood pressure and blood cholesterol based on an individual's absolute cardiovascular risk. *Lancet*. 2005;365:434–441.

TABLE 34.8 Recommendations of JNC 7 for Follow-up Based on Initial Blood Pressure Measurements for Adults

INITIAL BP (MM HG) ^a		
Systolic	Diastolic	Follow-up Recommendation
<130	<85	Recheck in 1 year
130–139	85–89	Recheck in 1 year; provide information about lifestyle modification
140–159	90–99	Confirm within 2 months
160–179	100–109	Evaluate or refer to source of care within 1 month
≥180	≥110	Evaluate or refer to source of care immediately or within 1 week, depending on clinical situation

^aIf systolic and diastolic categories are different, follow recommendations for the shorter time for follow-up. The schedule for follow-up should be modified according to reliable information about past BP measurements, other cardiovascular risk factors, or target organ disease. BP, Blood pressure; JNC, Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure. Modified from Chobanian AV, Bakris GL, Black HR, et al. Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure; National Heart, Lung, and Blood Institute; National High Blood Pressure Education Program Coordinating Committee. Seventh report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure. *Hypertension*. 2003;42:1206–1252.

rare when automated office BPs are taken with the patient alone in the examination room (as was recommended in the SPRINT trial³²). Although the most recent meta-analysis of 27 observational studies suggested that people with white coat hypertension have an increased long-term risk for all-cause and CV mortality (mostly driven by data from a large Spanish registry, which have since been retracted),³⁹ several prior studies (using rigorous diagnostic criteria) concluded that CV risk in untreated persons was not significantly elevated compared with normotensive individuals (Fig. 34.7). If CV risk is indeed elevated in white coat hypertension, it may be more appropriately attributed to subtle target organ damage and biomarkers that are intermediate

between normotensive and sustained hypertensive persons. Although many people with white coat hypertension will progress to sustained hypertension, most authorities recommend initial lifestyle modifications without drug therapy, ongoing risk assessment, and close follow-up.

Masked hypertension. Masked hypertension, defined as normal BP in the medical setting but elevated during ambulatory (or, perhaps less strictly, home) BP measurements, has a prevalence of 10% to 15% in the general population. Not surprisingly, such people often have target organ damage (which is often the only clue on initial evaluation) and a CV prognosis that is not much different than individuals with sustained hypertension⁴⁰ (see Fig. 34.7).

Sustained hypertension. Sustained (sometimes persistent) hypertension is diagnosed when the BP is elevated both inside and outside the medical setting, including at home and during usual daily activities. Patients with sustained hypertension should receive antihypertensive drug therapy, which reduces their risk for CV and renal events, albeit not to the level of normotensives (see Fig. 34.7).

Pseudohypertension. Pseudohypertension is defined as a significantly higher BP measured by cuff compared with that measured by an intraarterial catheter. Typically, this condition occurs in older patients and is attributed to calcium deposition, atheromatosis, and/or medial hypertrophy of the brachial (and likely other) arteries. It is suggested by the presence of a “positive Osler maneuver,” in which the nonperfused radial or brachial artery is still palpable after inflating the cuff to greater than 20 mm Hg higher than the palpated SBP. Confirmation requires arterial catheterization documenting a BP 10 to 15 mm Hg lower compared with a cuff-measured BP.

Isolated systolic hypertension. Isolated systolic hypertension (SBP ≥140 mm Hg, DBP <90 mm Hg) is recognized by current hypertension guidelines in Europe²⁶ but not in the United States. Its prevalence increases markedly with age, becoming by far the most common form of hypertension after the age of 60 years. It arises from vascular aging and large artery stiffening, both of which reduce their capacitance, accelerate pulse wave velocity, and widen the pulse pressure. Perhaps as a consequence of these processes, SBP continuously increases throughout life, whereas DBP usually decreases after the age of 50 years (Fig. 34.8).⁴¹ Although there was once great concern that lowering BP in patients with isolated systolic hypertension would push the DBP below the J-curve inflection point (and increase the risk for cardiac ischemia), all three RCTs (Systolic Hypertension in the Elderly

Cardiovascular Risk With Hypertension and Untreated Blood Pressure Patterns

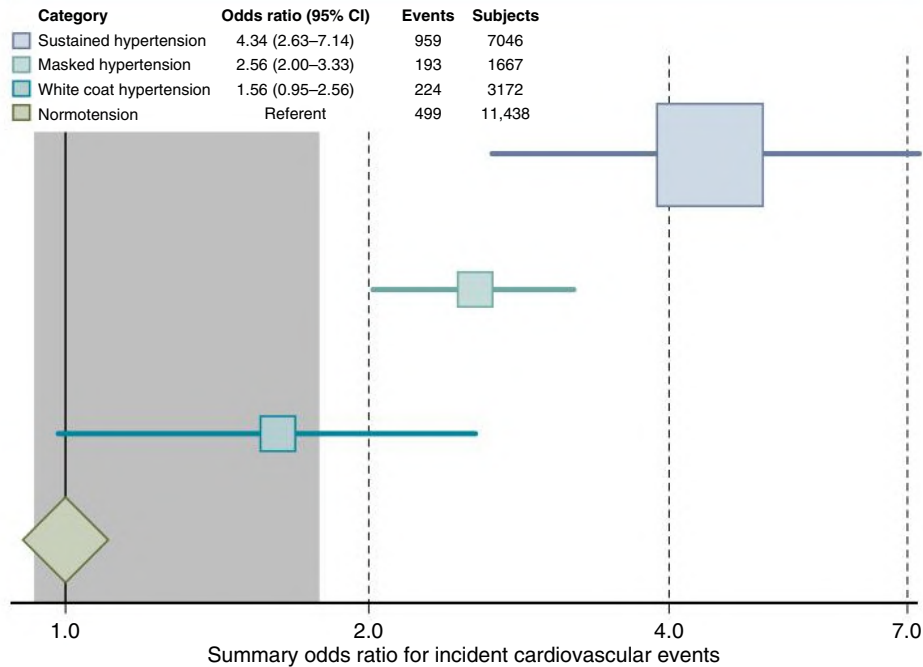


Fig. 34.7 Results of separate recent meta-analyses of various types of cardiovascular risk associated with treated and controlled hypertension and four types of untreated blood pressure patterns, as classified by ambulatory or home blood pressure monitoring: normotension, white coat hypertension, masked hypertension, and sustained hypertension. *Horizontal lines* represent the bounds of the 95% confidence intervals (CI), *squares* are drawn in proportion to the number of events. The cross-hatched area corresponds to the 95% confidence limits of the risk of treated and controlled hypertension (272 events/1829 subjects), indexed to masked hypertension (222 events/997 subjects⁴⁰). The many sources of inhomogeneity in these data (e.g., types of cardiovascular events, treated vs. untreated subjects, baseline demographics) are ignored in these comparisons. (Modified from Cohen JB, Lotito MJ, Trivedi UK, Denker MG, Cohen DL, Townsend RR. Cardiovascular events and mortality in white coat hypertension: a systematic review and meta-analysis. *Ann Intern Med.* 2019;170:853-862; and Pierdomenico SD, Pierdomenico AM, Coccina F, et al. Prognostic value of masked uncontrolled hypertension. *Hypertension.* 2018;72:862-869.)

Changes in Systolic and Diastolic Blood Pressure With Age

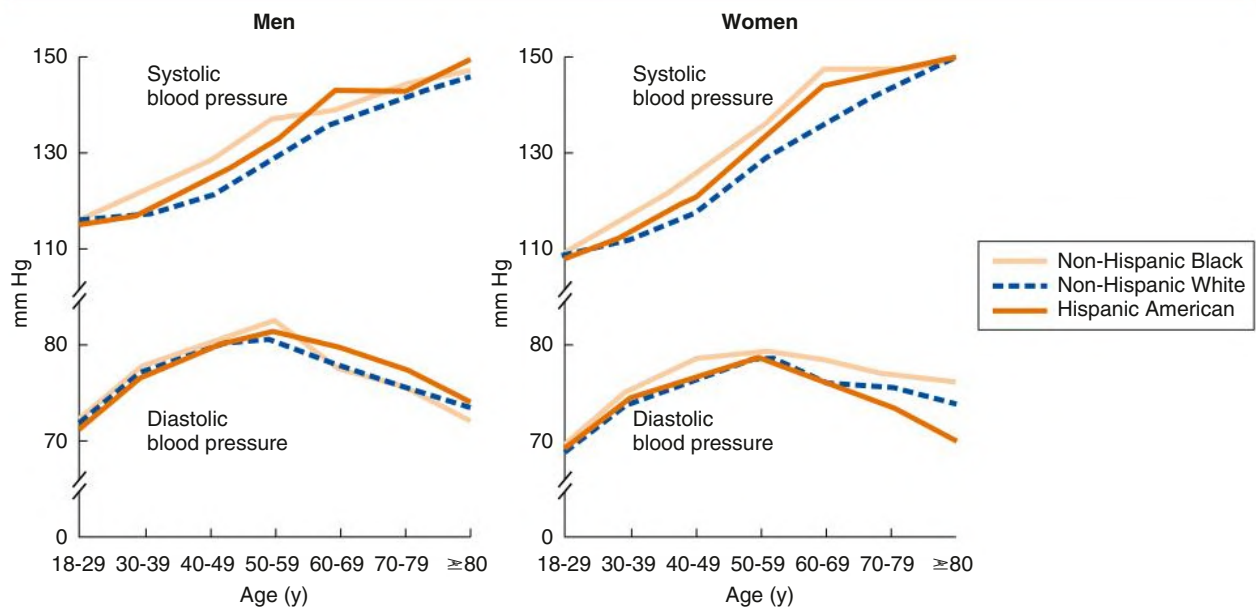


Fig. 34.8 Changes in Systolic and Diastolic Blood Pressures With Age. Systolic blood pressure and diastolic blood pressure by age and race or ethnicity for men and women older than 18 years in the US population (NHANES III, 1988–1991). (From Burt VL, Whelton P, Rocella EJ, et al. Prevalence of hypertension in the US adult population: results from the Third National Health and Nutrition Examination Survey, 1988–1991. *Hypertension.* 1995;23:305–313.)

Program in the United States, Systolic Hypertension in Europe trial, and Systolic Hypertension in China trial) showed a similar, significant decrease in stroke and no increase in cardiac events, with active drug treatment compared with placebo. The significant benefits on stroke and mortality of antihypertensive drug therapy (beginning with a thiazide-like diuretic) have now been extended to relatively healthy octogenarians (most of whom had isolated systolic hypertension) in the Hypertension in the Very Elderly Trial.⁴² This RCT established a SBP target of less than 150 mm Hg for octogenarians, which was also recommended by JNC 8 for patients older than 60 years,²⁵ but a dissenting minority of panel members later suggested that the more traditional SBP target of less than 140 mm Hg is reasonable, even if not directly supported by RCT evidence.⁴³ This position was strengthened by the subgroup analysis of SPRINT subjects older than 75 years, who had a 34% reduction in the primary composite end point and a 33% reduction in mortality, if randomized to the SBP of less than 120 mm Hg.⁴⁴ The 2021 KDIGO BP guidelines did not recommend a universal diastolic BP goal, primarily because of the lack of evidence in several important renal outcomes trials (e.g., Modification of Diet in Renal Disease, African American Study of Kidney Disease and Hypertension, both of which used mean arterial pressures, not systolic/diastolic BPs).³⁵

Resistant hypertension. Resistant hypertension is defined as BP above treatment goal, despite optimal doses of three antihypertensive drugs, including a diuretic.⁴⁵ Some clinical trials suggested that resistant hypertension may occur in 30% of hypertensive patients, but more population-based surveys put the US prevalence at 9% to 20%. Older age and obesity are strong risk factors; in several data sets from Southern European countries, white coat hypertension is found in up to 38% of individuals with resistant hypertension. Although nonadherence to prescribed antihypertensive drug therapy is the most common cause overall, suboptimal medication regimens and secondary hypertension (especially sleep apnea or hyperaldosteronism) are more frequently seen in referral centers.

Hypertensive emergencies and urgencies. A hypertensive emergency exists when elevated BP (usually BP >180/120 mm Hg) is associated with severe, ongoing target-organ damage (e.g., aortic dissection, hypertensive encephalopathy, cardiac ischemia, hematuria), which benefits from BP reduction within minutes to hours, typically using intravenous, short-acting antihypertensive agents in an intensive care unit (see Chapter 38). Examples include acute aortic dissection, acute decompensated heart failure, intracranial hemorrhage, eclampsia, and pheochromocytoma crisis.⁴⁶ “Hypertensive urgencies” often have similar elevations in BP but no ongoing severe target organ damage, and they can be handled in the outpatient department with oral antihypertensive agents and quick follow-up efforts. Aggressive drug therapy and intensive monitoring in this situation has little benefit.⁴⁷

Hypertension in children and adolescents. As discussed earlier, hypertension in youth was defined in the United States (until 2017) on a purely statistical basis: SBP or DBP at or above the 95th percentile for age, sex, and height, measured on at least three occasions. Although this made quality assurance for hypertension simple in large pediatric practices, the 2017 hypertension guidelines for children and adolescents have made this rubric more complex.²⁹ Children have a higher risk for secondary hypertension than adults; most secondary hypertension in young children has a renal origin. Most adolescents with hypertension have primary hypertension; risk factors include a family history of hypertension and obesity, but renal causes of hypertension are more common in stage 2 hypertension (see Table 34.9).

Hypertension in pregnancy. Hypertension occurs in more than 5% of all pregnancies and in approximately 5% of women taking oral contraceptives.⁴⁸ Definitions and implications of hypertension in pregnancy are discussed in Chapters 44 and 45.

TABLE 34.9 Secondary Hypertension: Acquired Causes

Condition/Disorder	Diseases/Comments
Kidney disorders	Kidney parenchymal disease, including acute and chronic glomerular diseases, chronic tubulointerstitial disease, PKD, diabetic nephropathy, and obstructive uropathy Renovascular disease: renal artery stenosis caused by atherosclerosis or fibromuscular dysplasia; arteritis; extrinsic compression of renal artery Other renal causes: renin-producing tumors, renal sodium retention (Liddle syndrome)
Endocrine disorders	Adrenocortical disorders: primary aldosteronism, congenital adrenal hyperplasia, Cushing syndrome Adrenomedullary tumors: pheochromocytoma (also extraadrenal chromaffin tumor) Thyroid disease: hyperthyroidism, hypothyroidism Hyperparathyroidism with hypercalcemia Acromegaly Carcinoid tumors
Exogenous substances and/or drugs	Oral contraceptives, sympathomimetics, glucocorticoids, mineralocorticoids, NSAIDs, calcineurin inhibitors, tyramine-containing foods and monoamine oxidase inhibitors, EPO, ergot alkaloids, amphetamines, herbal remedies, licorice (mimics primary aldosteronism), ethanol, cocaine and other illicit drugs, abrupt withdrawal of clonidine
Pregnancy	Preeclampsia and eclampsia
Coarctation of the aorta	Usually congenital, diminished pulses below the coarct, positive Hill's sign
Neurologic disorders	Sleep apnea Increased intracranial pressure: brain tumors Affective disorders Spinal cord injury: quadriplegia, paraplegia, Guillain-Barré syndrome Baroreflex dysregulation
Psychosocial factors	Hostility, time-urgency/impatience, depression, anxiety, occupational stress
Intravascular volume overload	Pedal or presacral edema, jugular venous distention, pulmonary rales
Systolic hypertension	Loss of elasticity of aorta and great vessels Hyperdynamic cardiac output: hyperthyroidism, aortic insufficiency, anemia, arteriovenous fistula, beriberi, Paget disease of bone
Obesity	White adipose tissue that has endocrine function: leptins, adiponectin, cytokines, chemokines, Ang II, other adipokines

Ang II, Angiotensin II; EPO, erythropoietin; NSAID, nonsteroidal antiinflammatory drug; PKD, polycystic kidney disease.

Classification by Cause of Hypertension

In 90% to 95% of adults with hypertension, the cause is unknown, and therefore the diagnosis is primary (or essential) hypertension (see Chapter 35). Table 34.9 lists the more common causes of secondary hypertension in descending order of prevalence in large

BOX 34.1 Guidelines for Measurement of Blood Pressure (BP): Patient Factors, Equipment, and Technique

Patient Factors

- Caffeine should not be consumed for 1 hour before the BP measurement.
- Cigarettes should not be smoked for at least 15 minutes before the BP reading.
- The standard BP measurement should be made with the patient not talking and seated comfortably, back and arm supported, and legs uncrossed. The cuff must be at the level of the heart, and the arm should be bare.
- The urinary bladder should be empty.
- Initially, BP should also be checked in both arms in the supine position after 5 minutes of rest; thereafter the arm with the higher reading is used for both supine and standing readings (after 2 minutes), especially in patients who are diabetic, older than 65 years, or receiving antihypertensive therapy. If sequential BP readings are taken in the same position, at least 30 seconds should elapse between BP readings. In patients younger than 30 years, check BP in one leg initially.
- To establish a diagnosis of hypertension, obtain BP readings on two different occasions, at least 1 week apart.

Equipment

- The bell of the stethoscope is preferred. The length of the bladder with the cuff should encircle at least 80% of the arm.
- The width of the cuff should be equal to two-thirds of the distance from the antecubital space to the axilla and should be 40% of the arm circumference. The best cuff for most adults is the 15-cm-wide cuff with a bladder of 33 to 35 cm in length. The distal edge of the cuff should be 2.5 cm (1 inch) above the antecubital fossa. For leg BP, thigh cuff length should encircle 80% of the thigh, and width should be 40% of the thigh circumference. For leg BP, the patient should be prone and popliteal artery sounds detected by auscultation.
- For infants, ultrasound equipment may need to be used.

Technique

- The initial systolic BP should be checked by palpating the disappearance of the radial or brachial pulse before auscultation and the cuff then deflated.
- The second BP check requires cuff inflation 20 mm Hg above the palpable systolic level.
- Deflate the cuff at a rate of 2 to 4 mm Hg per second for heart rates of 60 to 120, respectively.
- Record the Korotkoff sound I (appearance of sound) as the systolic BP and record the Korotkoff sound V (silence, 2 mm Hg below the last sound) as the more reproducible diastolic BP. If the sounds do not disappear, record the muffled sound (phase IV) as the diastolic BP.
- The sounds may be augmented by having the patient raise the arm and open and close the hand 10 times before inflating the BP cuff.
- Do not stop between systolic and diastolic BP readings; deflate the cuff, wait at least 30 seconds, and then reinflate. On each occasion, record at least two (and preferably three) BP readings. If the BP readings vary by more than 5 mm Hg, take additional BP readings until two are within 5 mm Hg.
- In children, the same standards apply for cuff size; Korotkoff sound V should be used. If the child is uncooperative, the systolic BP may be determined by palpation.

series. Renal parenchymal disease is most common (especially in nephrology clinics, particularly in diabetics; see [Chapters 31 and 33](#)), followed by endocrine causes ([Chapters 39 and 40](#)), renovascular hypertension ([Chapter 43](#)), sleep apnea, coarctation of the aorta, drug-induced hypertension, and rare monogenetic causes (see [Chapter 49](#)). Hypertension occurring in dialysis patients is covered in [Chapter 42](#), while neurogenic hypertension such as can occur with stroke is discussed in [Chapter 41](#).

TABLE 34.10 Acceptable Bladder Dimensions for Various Arm Sizes.

Usual Patient	Arm Circumference Range at Midpoint (cm)	Bladder Width (cm)	Bladder Length (cm)
Newborn	6	3	6
Infant ^a	6–15	5	15
Child ^a	16–21	8	21
Small adult	22–26	10	24
Adult	27–34	13	30
Large adult	35–44	16	38
Adult thigh	45–52	20	42

^aTo approximate a bladder width to arm circumference ratio of 0.4 more closely in children, additional cuffs are available. There is some overlap in the recommended ranges for arm circumference to limit the number of cuffs. It is suggested that the larger cuff be used if it is available. Modified from Pickering TG, Hall JE, Appel LJ, et al.

Recommendations for blood pressure measurement in humans and experimental animals, part 1: blood pressure in humans: a statement for professionals from the Subcommittee of Professional and Public Education of the American Heart Association Council on High Blood Pressure Research. Hypertension. 2005;45:142-161.

EVALUATION OF HYPERTENSION

Blood Pressure Measurement

Arterial BP is traditionally measured in the brachial artery with the cuff-based sphygmomanometer by detecting sounds that are generated (auscultatory method) or by recording vascular pulsations (oscillometric method) after decompression of a compressed artery.⁴⁹ [Box 34.1](#) lists guidelines for traditional BP measurement. The most common error is using an inappropriately sized cuff, because cuffs that are too small routinely underestimate the BP. [Table 34.10](#) provides acceptable bladder dimensions for varying arm sizes. Because of the low frequency of Korotkoff sounds, the bell of the stethoscope provides better detection than the diaphragm. Accurately taking BP is a teachable and testable skill, but many routine measurements are made with little effort to ensure their quality. As a result, many health care systems have transitioned to using automated oscillometric office BP measurements, which minimize the white coat effect, and after 5 minutes of quiet rest provide standardized triplicate measurements in a 5-minute period. Because elemental mercury (used in classic manometers and sphygmomanometers) has been eliminated from workplaces in most countries and aneroid (“dial”) manometers are fragile, oscillometric techniques have become increasingly popular. This method directly measures MAP and then uses a proprietary algorithm to estimate SBP and DBPs. Many such semiautomatic (typically for home BP readings) and fully automatic oscillometric devices have been validated against the traditional mercury column,⁵⁰ but they still require regular calibration, validation, and maintenance.

Variability of Blood Pressure

Office Versus Home Blood Pressure and Circadian Variation

BP varies considerably throughout the day and over time, even in the same individual. This intrinsic variation causes difficulty in identifying hypertensive individuals because there are many false negatives and even more false positives. Aside from measurement errors (which can be overcome by proper technique), intrinsic biologic variation can be addressed by taking at least three BP readings at a given visit. Recent US guidelines consider the lowest SBP and DBP at any given visit (even if not measured simultaneously) to be the BP for that visit.⁵¹ Confirming elevated

Increased Risk of Cardiovascular Mortality

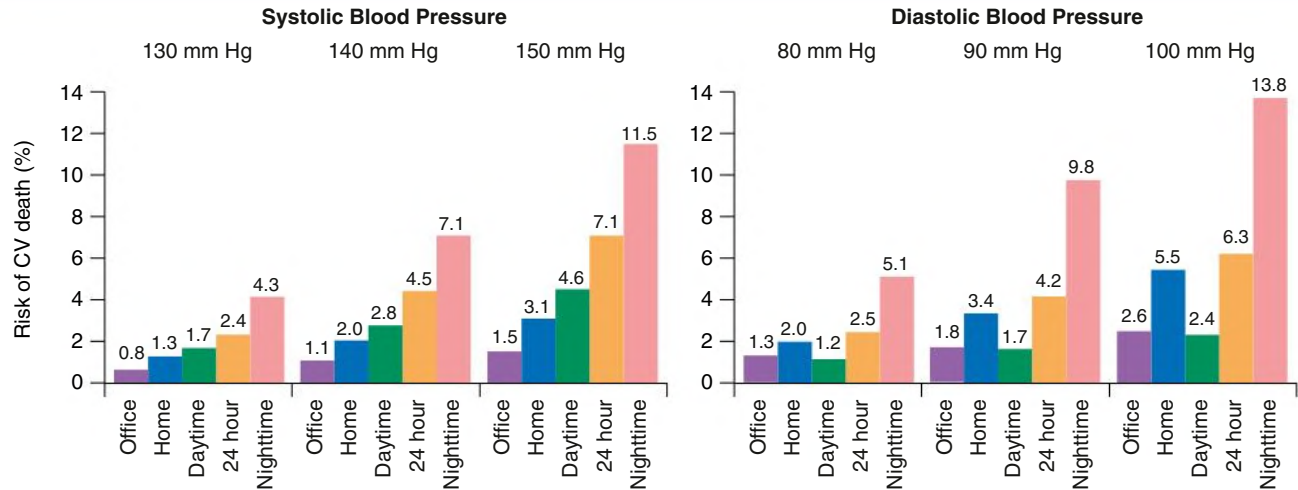


Fig. 34.9 Risk for Cardiovascular Mortality in an Early Population-Based Italian Study. Risk for cardiovascular death (%) over 11 years, for a 10 mm Hg increase in office, home, and ambulatory blood pressure (BP) readings at various initial BP values. CV, Cardiovascular. (From Segà R, Fachetti R, Bombelli M, et al. Prognostic value of ambulatory and home blood pressures compared with office blood pressure in the general population: follow-up results from the Pressioni Arteriose Monitorate e Loro Associazioni (PAMELA) study. *Circulation*. 2005;111:1777–1783.)

readings at a second visit takes advantage of regression to the mean to better identify the true BP for a given individual. Another widely recommended option is to measure home BPs (see later discussion).

In addition to physical/mental activity, and emotional/environmental stress, BP is affected by circadian variation. About 80% of the population display a dipping pattern, with average nocturnal BP that is 10% to 20% lower than the daytime average; this difference diminishes with increasing age. The normal morning surge in BP and heart rate that begins about 30 minutes before awakening correlates with an increased risk for death, myocardial infarction, and stroke in the early morning hours. Individuals with a nondipping nocturnal BP pattern more commonly have CKD, target-organ damage, secondary hypertension, autonomic dysfunction, and/or an increased risk for CV events.

Home and Ambulatory Blood Pressure Monitoring

Most ambulatory BP monitors use oscillometry and record BPs for 24 hours, making measurements frequently during the daytime (e.g., every 15 minutes) and less so at night (e.g., every 30 minutes). Inaccurate readings can occur with inappropriately sized cuffs, cardiac dysrhythmias, vigorous activity, inability to sleep because of pain resulting from cuff insufflation, and many other conditions. Because ambulatory BP monitoring (1) can diagnose white coat, masked, and nocturnal hypertension; (2) correlates better with both target-organ damage and future CV events than either office or home BPs; and (3) has been found to be cost-saving in many countries, it has been recommended (as at least one of several options) by many policymaking authorities, including British, European,²⁶ Canadian, and KDIGO Guidelines,³⁵ and the US Task Force on Preventive Services⁵² before starting antihypertensive drug therapy. Multiple national and international expert panels have issued guidelines regarding indications, technique, reporting, and implications of the procedure.^{34,49,53}

Because of the limited availability of, and scarce reimbursement for, ambulatory BP monitoring, home BP monitoring with semiautomated devices (typically duplicate measurements morning and evening for a week) has been recommended.^{26,34,49,53} Home BP readings, on average, are slightly higher than ambulatory BP readings (because the latter

Assessment of Patients for Hypertension by Use of Clinic, Home, and Ambulatory Monitoring

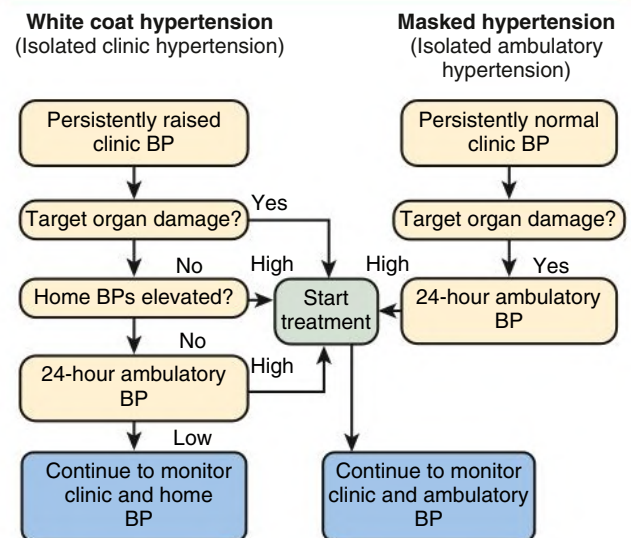


Fig. 34.10 Algorithm for diagnosis of hypertension using clinic, home, and/or ambulatory monitoring of blood pressure (BP).

includes nocturnal BPs) and slightly lower than office BPs. Home BP readings should be obtained with a device that has been calibrated (usually in the practitioner's office), are subject to reporting bias (which can be minimized by telemonitoring systems), and their interpretation is not often reimbursed. In general, home BP readings correlate better with target-organ damage and CV outcomes than office BP measurements but less well than ambulatory BP results (Fig. 34.9).⁵⁴ For many reasons, therefore, ambulatory BP monitoring is preferred over home BP recording in several settings⁴⁹ (Fig. 34.10 and Box 34.2).

BOX 34.2 Indications for Ambulatory Blood Pressure (BP) Measurement

- White coat hypertension
- Evaluation of apparent drug resistance
- Hypotensive symptoms (especially during sleep, e.g., in obstructive sleep apnea)
- Autonomic dysfunction
- Episodic hypertension
- Evaluation of nocturnal decreases in BP as a prognostic factor for target organ damage (e.g., left ventricular hypertrophy, ischemic optic neuropathy)
- Evaluation of BP changes in patients with paroxysmal nocturnal dyspnea and/or nocturnal angina
- Carotid sinus syncope
- Pacemaker syndromes
- Safety of withdrawing antihypertensive medication
- Assessment of 24-hour BP control in patient receiving once-daily medication
- Borderline hypertension with target organ damage (evaluation for masked hypertension)
- Evaluation of antihypertensive drug therapy in clinical trials

Risk Assessment in Hypertension

The medical history, physical examination, and a limited laboratory evaluation (biochemistry panel, including estimated GFR, and fasting glucose and lipid panel, urinalysis and urinary albumin-to-creatinine ratio, complete blood count, electrocardiogram) provide useful information to assess the presence and extent of target-organ damage and estimate future CV and renal risk (see [Chapter 35](#)).^{24-27,34} The ACC/AHA 2017 US hypertension guideline added serum thyroid-stimulating hormone to the list of recommended initial tests.³⁴

Consideration of Primary Versus Secondary Hypertension

If the history, physical examination, or screening laboratory studies suggest secondary hypertension, additional studies may be warranted ([Tables 34.11 and 34.12](#)). If renal parenchymal disease is suspected, ultrasound may be useful to evaluate renal size and echogenicity and rule out obstructive uropathy; Doppler flow studies also can help stratify risk for renovascular hypertension (see [Chapter 43](#)). Because primary aldosteronism can now be controlled either with chronic aldosterone antagonists or (occasionally) after laparoscopic surgery, a ratio of plasma aldosterone to plasma renin activity ratio may be useful (see [Chapter 39](#)). Suggested evaluations for other forms of secondary hypertension are listed in [Table 34.12](#).

Acknowledgment

I thank the late William J. Lawton, MD, and Drs. DiBona, Kopp, and Luft for their contributions to this chapter in previous editions.

TABLE 34.11 Evaluation for Primary Versus Secondary Hypertension

Classification	Medical History	Physical Examination	Laboratory Studies
General information and evaluation of target organs	Duration and course of hypertension Prior workup and treatment Diet/lifestyle: salt intake, tobacco, caffeine	Evaluation of volume status, optic fundi, heart, lungs, peripheral vessels, and nervous system	Complete blood count, fasting glucose, lipid profile (includes HDL, LDL, cholesterol, triglyceride), uric acid Consider echocardiogram
Primary (essential or idiopathic) or secondary (?)	Family history: Hypertension, cardiovascular and renal diseases Symptoms of target organ disease (related to eyes, central nervous system, cardiorespiratory, and peripheral vasculature)	See Table 34.12 for signs suggestive of secondary hypertension	See Table 34.12 for additional laboratory studies to rule out secondary hypertension

HDL, High-density lipoprotein; LDL, low-density lipoprotein.

TABLE 34.12 History, Physical Examination, and Initial Laboratory Evaluation for Secondary Hypertension

Target Organ/System	Medical History	Physical Examination	Laboratory Studies
Kidney			
Kidney parenchymal	History of kidney disease (including glomerulonephritis, nephrotic syndrome, calculi, urinary tract infection) Symptoms include nocturia, frequency, dysuria, hesitancy, urgency, incomplete emptying, dribbling, hematuria, pyuria, flank pain	Tenderness in costovertebral angles; palpable kidneys, edema	BUN, serum creatinine; urinalysis, urine culture if indicated; first morning UACR
Renovascular hypertension		Epigastric bruit; other vascular bruits	Kidney ultrasound with duplex Doppler flow study; consider angiography or magnetic resonance angiography

TABLE 34.12 History, Physical Examination, and Initial Laboratory Evaluation for Secondary Hypertension—cont'd

Target Organ/System	Medical History	Physical Examination	Laboratory Studies
Endocrine			
Primary aldosteronism	Muscle weakness, cramps		Serum aldosterone/plasma renin activity ratio
Cushing syndrome	Weight gain, muscle weakness, changes in body habitus	Body habitus: Central obesity, dorsocervical fat pad, abdominal striae	Midnight salivary cortisol; consider morning serum cortisol after dexamethasone suppression
Pheochromocytoma	Headaches, vasomotor symptoms, (inappropriate sweating, pallor), cardiac symptoms (awareness, tachycardia, palpitations)	Paroxysmal or intermittent hypertension (50% of patients)	24-hr urine for VMA, metanephrines, and catecholamines or plasma fractionated metanephrines
Carcinoid	Flushing		24-hour urine for 5-hydroxyindoleacetic acid
Hyperthyroidism	Weight loss, tachycardia, palpitations, sweating, heat intolerance	Palpable thyroid	Ultrasensitive serum thyroid-stimulating hormone level
Hypothyroidism	Weight gain, dry skin, cold intolerance, hair loss	Palpable thyroid	Serum ultrasensitive thyroid-stimulating hormone level
Hyperparathyroidism	Nausea, vomiting, bone pain, nephrolithiasis		Serum calcium, intact parathyroid hormone levels
Acromegaly	Change in size of head, hands, or feet (adult)	Appearance	Serum insulin-like growth factor-1 level (see Box 40.1)
Medication			
	Review of prescribed and over-the-counter medications (especially oral contraceptives, NSAIDs, sympathomimetic agents [cold and allergy remedies], illicit or recreational drugs, including alcohol, herbal remedies)		
Coarctation of the Aorta			
	Onset or detection of hypertension in childhood or adolescence	Simultaneous palpation of radial and femoral arteries to detect pulse lag in femoral arteries; leg blood pressure	Chest radiograph for heart size, configuration of aorta, rib notching; consider echocardiogram
Neurologic Disorders			
Sleep apnea	Obesity; weight gain; daytime somnolence; snoring, poor sleep habits (frequent awakening, not rested on arising); early-morning headache	Obesity, narrowed airway in hypopharynx, redundant pharyngeal tissue	Berlin questionnaire; consider formal sleep study (polysomnography)
Increased intracranial pressure	Headache, neurologic symptoms	Papilledema	Increased cerebrospinal fluid pressure
Affective disorders ^a			
Spinal cord injury ^a			
Psychosocial Factors			
	Family and support structure, occupation, education, stressors		
Volume Overload			
	Excess salt and water intake (may be iatrogenic with excess parenteral fluid)	Increased jugular venous distention, pulmonary crackles, presacral and peripheral edema, hepatomegaly	Chest radiograph
Isolated Systolic Hypertension			
		Pseudohypertension (positive Osler maneuver), cardiac and vascular examination (aortic insufficiency, arteriovenous fistula)	

^aMedical history, physical examination, and laboratory tests are beyond the scope of this discussion.

A more detailed discussion is provided in other relevant chapters. Pregnancy-associated hypertension is discussed in [Chapters 44 and 45](#). BUN, Blood urea nitrogen; NSAID, nonsteroidal antiinflammatory drug; UCAR, urine albumin-to-creatinine ratio; VMA, vanillylmandelic acid.

SELF-ASSESSMENT QUESTIONS

1. Short-term (within seconds to minutes) regulation of blood pressure in normal humans is most often attributed to which of the following physiologic systems?
 - A. Atrial natriuretic peptides
 - B. Endothelin
 - C. Kallikrein-kinin system
 - D. Renin-angiotensin-aldosterone system
 - E. Sympathetic nervous system
2. An asymptomatic, presumably healthy 67-year-old man visits the physician's office and has seated blood pressures of 140/96, 138/94, and 142/88 mm Hg, as measured personally and appropriately by the physician. According to guidelines promulgated by the Healthcare Effectiveness Data and Information Set (HEDIS) 2021 and the ACC/AHA 2017 US hypertension guideline, his blood pressure at this visit is most appropriately classified using which of the following terms?
 - A. Normal
 - B. Elevated
 - C. Stage 1 hypertension
 - D. Stage 2 hypertension
 - E. Isolated systolic hypertension
3. A 62-year-old White man visits the nephrologist because his routine tests last week showed that his estimated glomerular filtration rate was 59 mL/min/1.73 m² and his first morning voided urine contained 28 mg (normal <30) of albumin/gram of creatinine. Past medical history includes "prehypertension" since 2015, but he takes no medications. Blood pressures are 136/84, 134/86, and 132/82 mm Hg when measured personally and appropriately by the nephrologist. His 10-year risk of a cardiovascular event is estimated at 9.2%. According to the ACC/AHA 2017 US hypertension guideline, the most appropriate recommendation for this patient is which of the following?
 - A. Recheck blood pressure in this office in 1 year.
 - B. Recheck blood pressure in this office in 1 year, and provide guidance about lifestyle modifications.
 - C. Repeat blood and urine tests in 3 months, and recheck blood pressure in this office soon thereafter.
 - D. Recheck blood pressure within 2 months.
 - E. Institute antihypertensive drug therapy today with an angiotensin converting-enzyme inhibitor or angiotensin II receptor blocker.
4. A 32-year-old woman is referred to the nephrologist because her ophthalmologist noted grade II hypertensive retinopathy last month, and her obstetrician/gynecologist noted 2+ proteinuria on dipstick (without blood or nitrite), despite normal office blood pressures last week. Past medical history is remarkable for preeclampsia 4 and 6 years ago, but both of her children were delivered without complications, and her blood pressure has since been normal. Office blood pressures are 128/84, 126/82, and 124/80, measured by a calibrated automated oscillometric device. In addition to the retinopathy, electrocardiogram shows voltage criteria for left ventricular hypertrophy, and a random urinalysis shows 3+ protein. According to recent surveys, her condition is most commonly seen in what proportion of which population?
 - A. Less than 1% of people older than 65 years
 - B. 5% to 10% of the hypertensive population
 - C. 10% to 15% of the general population
 - D. 20% to 25% of the hypertensive population
 - E. 29% to 32% of the general population
5. A 56-year-old non-Black man is referred to the nephrologist because his blood pressures in his primary care medical home have never been below 190/120 mm Hg, yet the rest of his evaluation is normal. A request for prior authorization for ambulatory blood pressure monitoring was denied by his insurance plan. His home blood pressures, measured with a validated semiautomatic oscillometric device, have a mean and standard deviation of 128 ± 4/78 ± 3 (122 measurements over 3 months). He has no family history of hypertension. Office blood pressures, taken with the same machine, were 194/122, 196/120, and 192/118 mm Hg. Physical examination and laboratory studies were unremarkable. The most appropriate recommendation for him is which of the following?
 - A. A renal biopsy is indicated to identify the cause of his hypertension.
 - B. He should have an iothalamate clearance measured to assure that the estimated glomerular filtration rate is accurate.
 - C. He should undergo renal ultrasound with Doppler flow studies, a 24-hour urine collection for vanillylmandelic acid and metanephrines, and a plasma aldosterone/renin activity ratio to rule out common causes of secondary hypertension.
 - D. He should start an angiotensin converting-enzyme inhibitor or an angiotensin II receptor blocker today and return to this office in 1 month for repeat blood pressure measurement by his home device.
 - E. He should continue to monitor his home blood pressures and call for another appointment when/if they meet or exceed 125/75 mm Hg.

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